

IV. INTEGRATION POSSIBILITIES AND EXPERIMENTAL RESULTS

Although this circuit was first designed as a discrete component device, one may consider the possibility of integrating it into a monolithic opamp.

1) Only two components (a single transistor and a single resistor) have to be added into the IC design.

2) The transistor may be either of type n-p-n or of type p-n-p without affecting the working.

3) The same is true for the Si or Ge nature, which has no influence.

4) The integrated resistor R_1 is in the range from 1 to 10 k Ω and easy to deposit.

5) The external potentiometer is connected as a variable resistor between one power supply and a control pin. Its value is in the kilohm range while keeping the low-voltage source impedance near 1 Ω .

Therefore, the whole system can be built with a standard high-yield silicon IC process and needs only a single extra pin for control purpose.

Presently our laboratory has no facilities for making IC's. We have only tested discrete components applications, several of which have been used quite satisfactorily for more than a year. Some experimental results are given in Table I with V_1 and V_2 equal to ± 15 V according to transistor polarity, and $R_1 = 2$ k Ω . They are in good agreement with theoretical predictions.

REFERENCES

- [1] R. D. Middlebrook, *Differential Amplifiers*. New York: Wiley, 1963, chs. 3 and 5.
- [2] G. Fontaine and G. Reboul, "Automatic gain control in signal processing for bubble chamber pictures," in *Proc. Int. Conf. Data Handling Systems in High Energy Physics*, Cambridge, England, Mar. 1970; Reprinted by CERN (Geneva) CERN 70-21, pp. 383-398.
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Comment on "Single-Ended-Input Single-Ended-Output Four-Quadrant Analog Multiplier"

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In the above special correspondence,¹ Scratchley comments on the unipolar limitations of the product-quotient circuit Fig. 1 of Scratchley (see [1]) and then proceeds to describe a means of using two such circuits to generate a "transposed origin" four-quadrant product (Fig. 2 of Scratchley).

A much simpler solution was published in a companion paper to that in which the product-quotient circuit was published. The paper [2] described what is now a widely used four-quadrant multiplier, and several of the circuits discussed

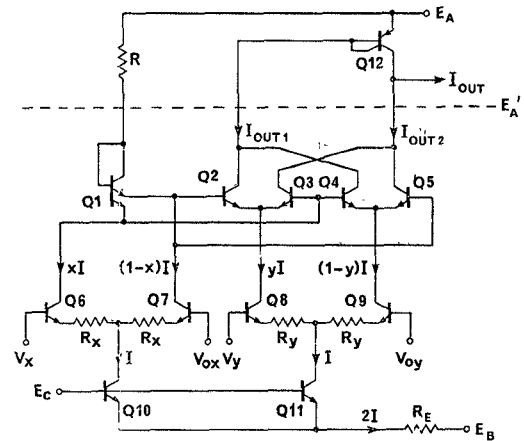


Fig. 1

in that paper can produce an offset four-quadrant product with about half the number of transistors required in Fig. 2 of Scratchley and suffer from fewer matching requirements.

Fig. 1 shows one alternative. In [1] and [2] the same supply notations are used but Q15 is omitted, which served little purpose and reduced the output capability by about 2 V. It is simply shown that,

$$I_{OUT1} = \frac{I}{2}(1 + XY) \quad (1)$$

and

$$I_{OUT2} = \frac{I}{2}(1 - XY), \quad (2)$$

where

$$X = (2x - 1) = \frac{V_x}{IR_x} \quad (3)$$

and

$$Y = (2y - 1) = \frac{V_y}{IR_y}. \quad (4)$$

This result was given in [2, footnote 1].

Since the requirement is for single-sided inputs, input offsets can be corrected on the unused inputs to Q7 and Q9. Typically, the offset voltage will be about 1-5 percent of IR_x or IR_y , depending primarily on the matching of V_{BE} in Q1-Q5. Separate transistors can of course be used to realize the dual-emitter transistor Q1.

A single-ended output can be simply derived using a dual collector lateral p-n-p transistor Q12. Assuming unity gain for this device,

$$I_{OUT} = IXY. \quad (5)$$

The resistor R provides a small bias voltage for Q2-Q5; the product IR should be about 200 mV.

REFERENCES

- [1] B. Gilbert, "A new wide-band amplifier technique," *IEEE J. Solid-State Circuits*, vol. SC-3, pp. 353-365, Dec. 1968.
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Manuscript received February 7, 1972; revised April 3, 1972.
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¹ E. D. Scratchley, *IEEE J. Solid-State Circuits*, vol. SC-6, pp. 394-395, Dec. 1972.